[VIT logo]

OSmosis

A course-grounded tutor for Operating Systems: retrieval-augmented explanations, exact numerical solvers and diagnosis of student reasoning

[Course code] - [Course name]

[Student name] | [Registration number]

Faculty: [Faculty name]

[School / Department], Vellore Institute of Technology | [Academic year]

Introduction and objectives

The problem

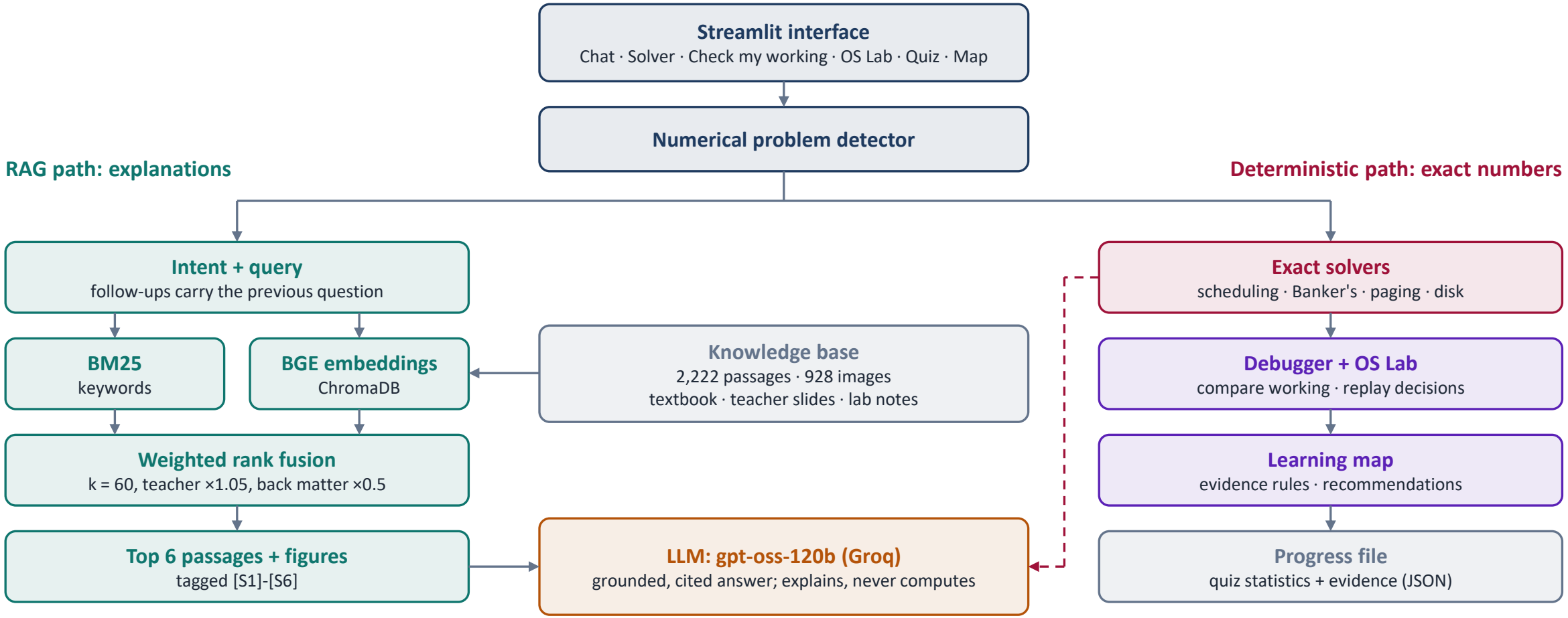
- **Static material shows results, not reasoning:** a worked example gives the final schedule, not why each decision was taken.
- **Numerical problems are error-prone:** one missed preemption changes every later value, and final answers cannot show where it went wrong.
- **Generic AI answers are not tied to the course:** different conventions, no citations, unreliable arithmetic.

Goal: explanations grounded in the course material, exact answers to numerical problems, and feedback on the step where a student's reasoning fails.

Objectives and status

1	Course-grounded answers with RAG	Achieved
2	Citations, figures and slide images	Achieved
3	Exact solvers for four problem families	Achieved + tested
4	Step-level diagnosis of student working	Achieved + tested
5	Interactive algorithm execution (OS Lab)	Achieved + tested
6	Misconception tracking, targeted practice	Rules not validated
7	Quiz, viva and adaptive difficulty	LLM output not evaluated
8	Retrieval evaluation and automated tests	Achieved

Methodology: two paths, one interface



Dashed: the solver's exact result is passed to the LLM as fixed text to explain. Without an API key, every feature except written answers still works.

Stack Python 3.12 · Streamlit 1.65 · rank_bm25 · BAAI/bge-base-en-v1.5 (sentence-transformers) · ChromaDB 1.5.9 · gpt-oss-120b via Groq (OpenAI-compatible API) · PyMuPDF · python-pptx

R&D: tuning retrieval and constraining the LLM

Retrieval experiments

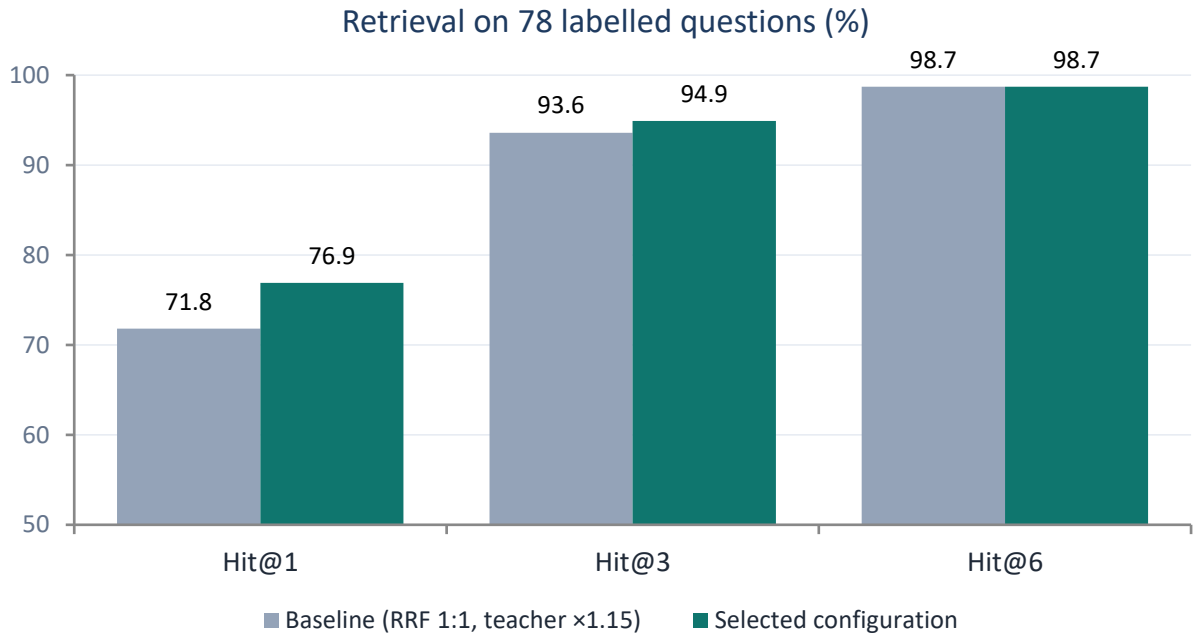
78 labelled questions; 19 held out. A grid of 147 fusion settings was scored on the other 59; the best MRR that did not lower Hit@3 or Hit@6 was selected.

Prompt engineering

Answer only from the numbered SOURCES, cite every factual sentence [S#], say when the sources do not cover it. Invalid citation tags are stripped in code; quiz JSON is validated.

Implementation

Solvers record every decision; the debugger, OS Lab and learning map read those states. 71 automated tests in 7 suites, all passing.



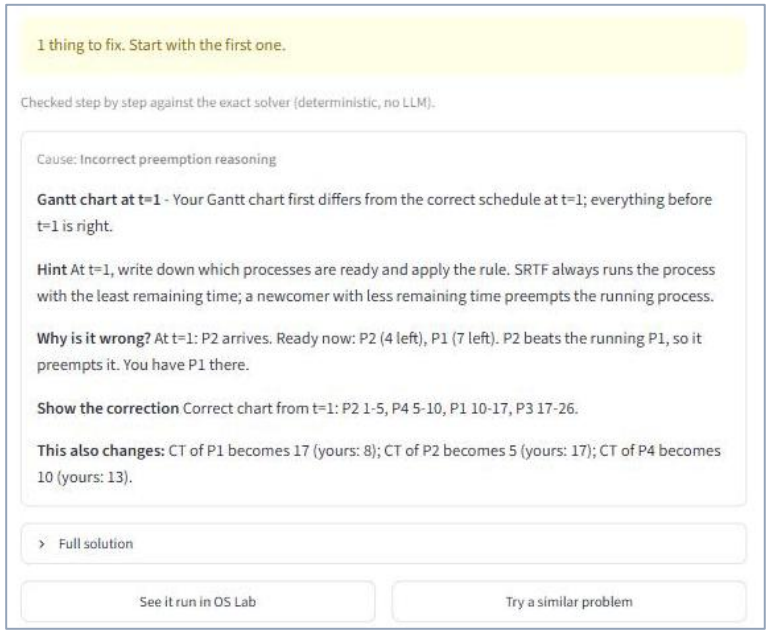
71.8% → 76.9%
Hit@1, all 78 questions

84.2% → 94.7%
Hit@1, 19 held-out questions

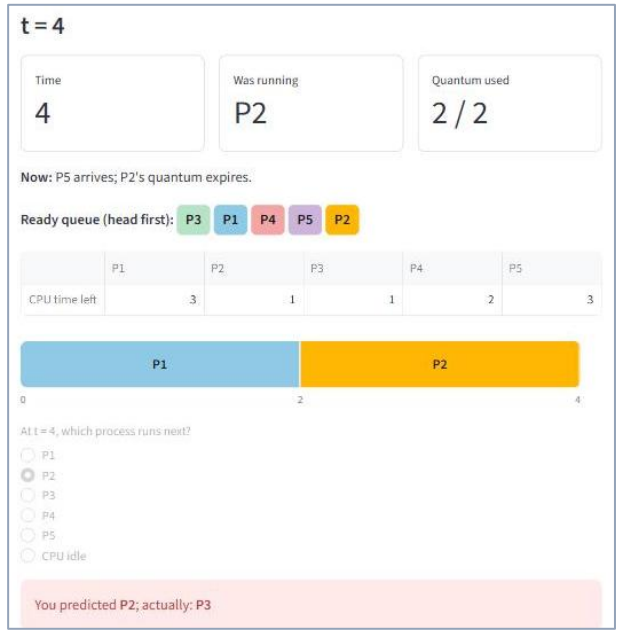
0.830 → 0.858
MRR, all 78 questions

Hit@6 is unchanged at 98.7%. Per-topic changes are not statistically significant: the smallest topics have only 3 and 5 questions.

Features in use, and how they were checked



Check my working: a missed SRTF preemption found at t = 1, with cause, hint and correction



OS Lab: Round Robin at t = 4; the student predicted P2, the scheduler runs P3



Numerical solver: Banker's algorithm safe sequence with Work after each step

71 / 71
automated tests pass (7 suites), no API key needed

20
worked examples from the slides and textbook, solved in code in the chat

6 pages
exercised in the browser: chat, solvers, debugger, lab, map, quiz

Not yet
evaluated: LLM-written answers and questions; no student study

Conclusion and future work

What OSmosis shows

- Grounded explanation and exact computation are kept apart: the LLM explains from cited course passages, code produces every number.
- The same recorded solver states let the system diagnose a student's first wrong step and replay the real algorithm.
- Measured: Hit@1 71.8% → 76.9% (held-out 84.2% → 94.7%) with Hit@6 at 98.7%; 71 automated tests pass.

Future work

1. **Reranking stage**
   A cross-encoder over the fused candidates, to fix the adjacent-section errors that remain.
2. **Validation with students**
   A usability study, and checking the misconception rules against real outcomes.
3. **Evaluating LLM output**
   Measure answer and question quality, and make the progress file safe for concurrent sessions.

Not validated in this project: the quality of LLM-written content, and learning outcomes with students.

References

[1] A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 10th ed. Hoboken, NJ, USA: Wiley, 2018.

[2] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020, pp. 9459-9474.

[3] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333-389, 2009, doi: 10.1561/1500000019.

[5] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff, D. Lian, and J.-Y. Nie, "C-Pack: Packed resources for general Chinese embeddings," arXiv:2309.07597, 2023.

[9] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, 2009, pp. 758-759, doi: 10.1145/1571941.1572114.

[10] OpenAI, "gpt-oss-120b & gpt-oss-20b model card," arXiv:2508.10925, 2025.

Numbers follow the project report's reference list (16 entries), which also cites the libraries, model documentation and course material used.